Utsav Bandyopadhyay Maulik

Curriculum Vitae

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in *Utsav Bandyopadhyay Maulik*
Utsav Bandyopadhyay Maulik
Date of Birth: 28 June 2000
Nationality: Indian



Academic Background

Jan 2026 – Jan 2030 **PhD, School of Computing**, *National University of Singapore*, Singapore
- Supervisor: Prof. Jin Song Dong
- Research Area: Computer Vision, Video Understanding, Sports Analytics, Vision Transformers
- Current CGPA: 4.50

Jan 2023 – Jul 2025 **MS (by Research), Computer Science and Engineering**, *Indian Institute of Technology Kharagpur*, Kharagpur, West Bengal
- Coursework CGPA: 9.87/10.00
- Subjects: Machine Learning [Ex], Advanced Machine Learning [Ex], Natural Language Processing [B], Deep Learning [B], English for Technical Writing [Ex]
- Comprehensive Viva Exam: [Ex]; Thesis Defence: [Ex]
- **Thesis:** Enhancing Medical Domain Speech Recognition with Large Language Models - Grade: [Ex]

2018 – 2022 **Bachelor of Technology, Computer Science and Engineering**, *Techno India Saltlake, MAKAUT*, Kolkata, West Bengal
- Coursework CGPA: 8.82/10.00
- Final Year Project: Deep Learning based Pollution Detection in Vehicles

2018 **XII (ISC), Science [PCM] with Computer**, *St. James School*, Kolkata, West Bengal

2016 **X (ICSE), Science [PCM] with Computer**, *St. James School*, Kolkata, West Bengal

Work Experience

June 2022 – June 2024 **Junior Research Fellow**, in the project *Connected Numeral Recognition in Indian Language Speech* sponsored by Hitachi India Private Limited, IIT Kharagpur, West Bengal, India
- Responsibilities included data collection, preprocessing, model design, and evaluation for Indian language ASR.

January 2024 – June 2024 **Teaching Assistant [Programming and Data Strcture Lab]**, *IIT Kharagpur*, Kharagpur, West Bengal
- Took lab sessions, mentored B.Tech students & conducted examinations.

Aug 2025 - Dec 2025 **Researcher**, *TCS Research*, India
- Ranked Second in the 4-week Initial research challenge project. Worked on Time-Series data from smart wearables to detect stress in drivers and office workers.
- Current Project: AI for Plasmonic Sensing

Research Experience

June – Aug 2018 Machine Leraning Research Intern, Machine Intelligence Unit, Indian Statistical Institute, Kolkata

Sept – Dec 2021 INAE Summer Intern, "Determining effect of medications on a patient using week-wise radiomic features on CBCT images", under Prof. Jayanta Mukhopadhyay, IIT Kharagpur

2021 – 2022 B.Tech Final Year Project, "Vehicle pollution detection using Machine Learning and Deep Learning algorithms", Techno Main SaltLake, Kolkata

June 2024 Lab Visit, Laboratoire de Biopharmacie et Pharmacie Clinique, University of Rennes, France (Drug Data Collection & Data Cleaning from French Hospital)

Nov 2024 Lab Visit, Istituto Superiore di Sanità (ISS), Rome, Italy. Delivered talk: "Enhancing Domain Specific ASR Performance using Fine-Tuning and LLM Post-Processing: A Study in the Medical Domain."

2023 – 2025 M.Sc. By Research - "Domain Adaptation for Automatic Speech Recognition" under Prof. Pabitra Mitra & Prof. Sudeshna Sarkar, IIT Kharagpur

List of Publications

Journal Papers

1. R. Das, U.B. Maulik, B. Boote, S. Sen, & S. Bhattacharya, "Multi-path Convolutional Neural Network to Identify Tumorous Sub-classes for Breast Tissue from Histopathological Images" *SN Computer Science*, vol. 3, art. no. 363, 2022, DOI: 10.1007/s42979-022-01273-z67

Conference Papers

1. S. Kundu, U.B. Maulik, A. Bej, & U. Maulik (2020). *Deep Learning based Pollution Detection in Intelligent Transportation Systems*. In *Proceedings of the 2020 IEEE 5th International Conference on Computing, Communication and Automation (ICCCA)*, Greater Noida, India (pp. 292297). DOI: 10.1109/ICCCA49541.2020.9250883
2. R. Das, P. Basak, U. B. Maulik, S. Bhattacharya, & U. Maulik (2020). *A MultiStage GAN Based Framework for the Generation of High Resolution Mammography Images*. In *Contemporary Issues in Computing (ICCIC), Topics in Intelligent Computing and Industry Design*, India (pp. 7275). DOI: 10.26480/cic.01.2020.72.75
3. U.B. Maulik, P. Mitra, & S. Sarkar (2024). *Enhancing DomainSpecific ASR Performance Using Finetuning and ZeroShot Prompting: A Study in the Medical Domain*. In *Proceedings of the Sixth Doctoral Symposium on Intelligence Enabled Research (DoSIER 2024)*, Jalpaiguri, India (pp. 172180). Published as a workshop paper in CEUR Workshop Proceedings (ISSN: 16130073).

Technial Skills

Programming Languages	Python, C, Java, PL/SQL
Data Analysis	Pandas, NumPy, Matplotlib, Excel
ML Tools & Libraries	PyTorch, Scikit-learn, Hugging Face Transformers, OpenCV, TensorFlow
ML Techniques	LLM, RAG, CNN, Transformers, Transfer Learning, Fine-Tuning
Development Tools	VS Code, Jupyter Notebooks, Git
Soft Skills	Leadership, Teamwork, Communication, Presentation, Adaptability, Continuous Learning

Research Interests

Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Generative AI, Agentic AI, Large Language Models, Explainable AI

Conferences & Certifications

NPTEL & Coursera
- Design and Analysis of Algorithms (8 weeks, with exam)
- Computer Architecture (12 weeks, with exam)
- Neural Networks and Deep Learning

FDP One-week Faculty Development Program (FDP) on Optimization and Deep Learning organized by Electronica & ICT Academy, NIT Patna, supported by MeitY, Govt. of India (Excellent grade).

INDICON 2021 Participated at IIT Guwahati, 19 to 21 December 2021.

Additional Information

Competitive Exams
Qualified GATE 2022, CAT 2021 & WBJEE 2018

Language Proficiency
TOEFL: 110/120 (2022)

Extracurricular Activities
- Club-level cricketer for Cricket Association of Bengal, Howrah District.
- Inter-hall/inter-department cricket, football, and badminton player at IIT KGP.
- Certified rock climber.
- Volunteer, Zomato Feeding India under PANKH NGO Foundation.
- Youtuber at Utsav Explores, Videographer and Video Editor
- Secured 100 points in MAR during B.Tech.

Hobbies
Playing and watching sports, Travelling, Learning new technologies, Vlogging, Photo & Video editing (Adobe Premiere Pro, Edits, Lightroom & Canva), Photography, Singing, Watching movies & series

Countries Visited
USA, United Kingdom, China, Germany, France, Netherlands, Italy, Spain, Portugal, Switzerland, Slovenia, Croatia, Luxembourg, Thailand, Singapore, United Arab Emirates, Nepal, Bangladesh